

REACTION IN CSTR

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Objective:

The aim of this experiment is to analyze the performance of CSTR for second-order saponification reaction between Sodium hydroxide ($NaOH$) and ethyl acetate ($CH_3COOC_2H_5$) under steady-state conditions.

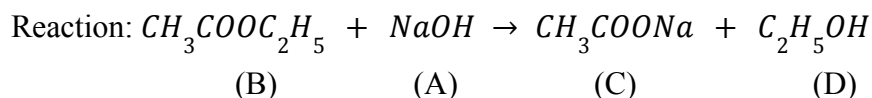
Theory and Background:

A Continuous Stirred Tank Reactor (CSTR) is a widely used reactor in chemical and biochemical industries for conducting homogeneous liquid-phase and gaseous-phase reactions. It operates under steady-state conditions, where reactants are continuously fed, and products are continuously removed, ensuring a uniform reaction environment. Constant stirring maintains homogeneous composition and temperature, making CSTRs particularly effective for processes requiring precise control over reaction conditions, such as polymerization, fermentation, and wastewater treatment.

In this experiment, we analyze the saponification reaction, a second-order reaction between sodium hydroxide ($NaOH$) and ethyl acetate ($CH_3COOC_2H_5$), forming sodium acetate (CH_3COONa) and ethanol (C_2H_5OH). The progress of the reaction can be monitored using conductivity measurements, as $NaOH$, a strong electrolyte, depletes over time, reducing the conductivity of the solution. This experiment helps in understanding reaction kinetics and the role of CSTRs in continuous production processes.

CSTR: It is a type of reactor in which the reactants are continuously fed, while constantly stirring the reaction mixture to maintain homogeneity.

Saponification reaction: A reaction where an ester reacts with base to form alcohol and salt. It is a second order reaction.



V_1 represents the volume of $NaOH$

V_2 represents the volume of Ethyl Acetate.

$$\text{Total Volume}(V) = V_1 + V_2$$

A is $NaOH$ and B is Ethyl Acetate

For CSTR, the mass balance is given by:

(Mass balance on A)

$$\text{Input} = \text{Output} + \text{Disappearance by reaction} + \text{Accumulation}$$

Input of A, moles/time = $F_{A0} (1 - X_{A0}) = F_{A0}$

Output of A, moles/time = $F_A = F_{A0}(1 - X_A)$

Disappearance of A by reaction moles/time = $(-r_A)V$

$$\Rightarrow F_{A0} - F_A = -r_A V$$

$$F_{A0} X_A = -r_A V$$

Where F_{A0} is the molar flow rate of reactant A entering (mol/s)

F_A is the molar flow rate of reactant A leaving (mol/s)

X_A is the conversion

$-r_A$ is the rate of reaction (mol/ L s)

V is the volume of reactor (L)

$$\text{Therefore, } X_A = \frac{-r_A V}{F_{A0}}$$

The rate law for the equation will be: $-r = kC_A C_B$

Since this is a reversible reaction, therefore the equation would be:

$$-r = k \left[C_A C_B - \frac{C_C C_D}{K_e} \right]$$

Where, $C_A = C_{A0}(1 - X_A)$, $C_B = C_{A0}(\theta - X_A)$, $C_C = C_{A0}X_A$, $C_D = C_{A0}X_A$

$$\text{On simplifying and substituting } \frac{V}{F_{A0}} = \frac{X_A}{kC_{A0}^2 \left[(1-X_A)(\theta-X_A) - \frac{X_A^2}{K_e} \right]}$$

Apparatus:

1. 0.05 N NaOH solution
2. 0.05 N ethyl acetate solution
3. Containers for reagents and product
4. Magnetic stirrer
5. Conductivity cell
6. Stopwatch
7. Graduated cylinder

Procedure:

1. Calibrate the conductivity cell.
2. Prepare 2 liters of 0.05 N NaOH solution and 0.05 N ethyl acetate solution.
3. Transfer the solutions into two separate overhead storage tanks.

4. Measure and adjust the flow rates of NaOH and ethyl acetate, ensuring that NaOH is the limiting reactant $v_{NaOH} < v_{EA}$.
 - a. Limiting reagent: The reactant which is completely used, and limits the extent of the reaction.
5. Begin measuring the conductivity over time.
6. Note the time when the flow becomes stable.
7. Continue recording conductivity until the readings remain nearly constant for about 2 minutes.
8. Once the conductivity stabilizes, stop the flow of both solutions and turn off the magnetic stirrer.

Known parameters:

- $v_{NaOH} = \frac{\text{volume filled}}{\text{time taken}} = \frac{10 \text{ ml}}{10 \text{ s}} = 1 \frac{\text{ml}}{\text{s}}$
- $v_{EA} = \frac{\text{volume filled}}{\text{time taken}} = \frac{10 \text{ ml}}{5 \text{ s}} = 2 \frac{\text{ml}}{\text{s}}$
- $k = 0.0632 \frac{\text{l}}{\text{mol s}}$
- $\text{Volume of CSTR} = 500 \text{ ml}$
- $\text{Initial Concentration of NaOH } (C_{A0}) = 0.01667 \frac{\text{mol}}{\text{ml}}$
- $\text{Initial Concentration of Ethyl Acetate } (C_{B0}) = 0.03333 \frac{\text{mol}}{\text{ml}}$
- $\text{Molar flow rate of NaOH } (F_{A0}) = 0.01667 \frac{\text{mol}}{\text{s}}$
- $\text{Molar flow rate of Ethyl Acetate } (F_{B0}) = 0.06667 \frac{\text{mol}}{\text{s}}$

Formulae and Calculation:

- *Calculation of initial Concentrations:*
 - $C_{A0} = \frac{0.05 * v_{NaOH}}{v_{NaOH} + v_{CH_3COOC_2H_5}} = 0.01667 \frac{\text{mol}}{\text{ml}}$
 - $C_{B0} = \frac{0.05 * v_{CH_3COOC_2H_5}}{v_{NaOH} + v_{CH_3COOC_2H_5}} = 0.03333 \frac{\text{mol}}{\text{ml}}$
- $F_{A0} = C_{A0} V$
 - $F_{A0} = C_{A0} \times v_{NaOH} = 0.01667 \frac{\text{mol}}{\text{s}}$
 - $F_{B0} = C_{B0} \times v_{NaOH} = 0.06667 \frac{\text{mol}}{\text{s}}$

- $$\frac{V}{F_{A0}} = \frac{X_A}{kC_{A0}^2[(\theta - X_A)(1 - X_A) - \frac{X_A^2}{K_e}]}$$

- Substituting $K_e = 1$ and $\theta = 1$, the equation simplifies to:

- $$\frac{V}{F_{A0}} = \frac{X_A}{kC_{A0}^2[(1 - X_A)(1 - X_A) - \frac{X_A^2}{1}]}$$

Thus, the final expression for conversion X_A is:

- $$X_A = \frac{V k C_{A0}^2}{F_{A0} + 2V k C_{A0}^2}$$

The value of theoretical conversion thus obtained is $X_A = 0.2564 = 25.64\%$

- $$X_A = 1 - \frac{N_{A0}}{N_A}$$
 where N_{A0} and N_A are the normality values

- This gives us experimental N_A maximum to be $0.2694 = 26.94\%$

There is slight variation in the theoretical values and the experimental values, this can be due to the fluctuation in the reading of the conductometer, human error and because of improper cleaning beaker before use.

Readings:

TABLE 1: TABLE CONTAINING CONDUCTIVITY AND CONCENTRATION DATA OF NaOH

Concentration (M)	Conductivity (mS)
0.05	7.07
0.025	3.503
0.0125	1.87
0.00625	0.9407
0.003125	0.4738
0.0015625	0.2617

TABLE 2: TABLE CONTAINING EXPERIMENTAL DATA

Time (s)	Conductivity (mS)
0	2.375
15	2.355
30	2.33
45	2.245

60	2.288
75	2.24
90	2.197
105	2.134
120	2.093
135	2.057
150	2.019
165	2.013
180	1.973
195	1.931
210	1.894
225	1.847
240	1.822
255	1.729
270	1.75
285	1.752
300	1.753
315	1.75

TABLE 3: TABLE CONTAINING CALCULATED DATA

Time (s)	Conductivity (mS)	Normality (N)	Conversion (Xa)	Concentration (M)
0.00000	2.37500	0.01657	0.00000	0.01667
15.00000	2.35500	0.01643	0.00862	0.01652
30.00000	2.33000	0.01625	0.01940	0.01634
45.00000	2.24500	0.01564	0.05604	0.01573
60.00000	2.28800	0.01595	0.03750	0.01604
75.00000	2.24000	0.01561	0.05819	0.01570
90.00000	2.19700	0.01530	0.07673	0.01539
105.00000	2.13400	0.01485	0.10389	0.01494
120.00000	2.09300	0.01456	0.12156	0.01464

135.00000	2.05700	0.01430	0.13708	0.01438
150.00000	2.01900	0.01403	0.15346	0.01411
165.00000	2.01300	0.01398	0.15605	0.01407
180.00000	1.97300	0.01370	0.17329	0.01378
195.00000	1.93100	0.01340	0.19140	0.01348
210.00000	1.89400	0.01313	0.20735	0.01321
225.00000	1.84700	0.01280	0.22761	0.01287
240.00000	1.82200	0.01262	0.23838	0.01269
255.00000	1.72900	0.01196	0.27847	0.01203
270.00000	1.75000	0.01211	0.26942	0.01218
285.00000	1.75200	0.01212	0.26856	0.01219
300.00000	1.75300	0.01213	0.26813	0.01220
315.00000	1.75000	0.01211	0.26942	0.01218

Results and discussion:

Conductivity (mS) vs. Concentration (M)

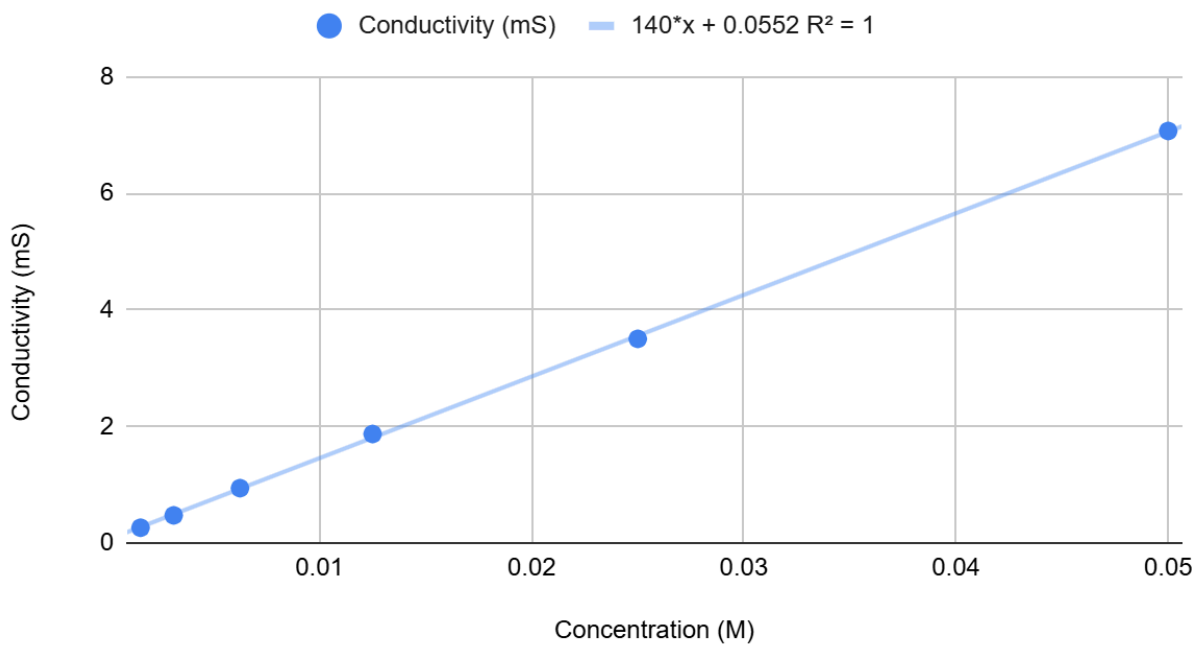


Figure 1: Figure showing the Conductivity(mS) vs Concentration (M) curve (Calibration curve)

Conductivity (mS) vs. Time (s)

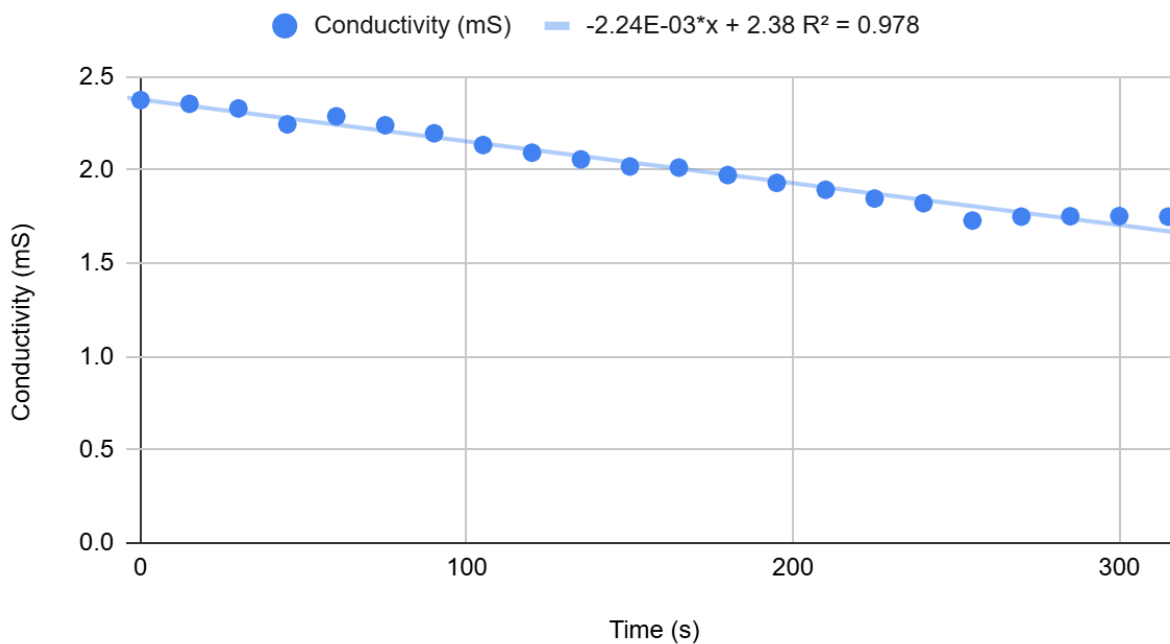


Figure 2: Figure showing the plot between Conductivity(mS) and Time (s)

Conductivity (mS) vs. Normality (N)

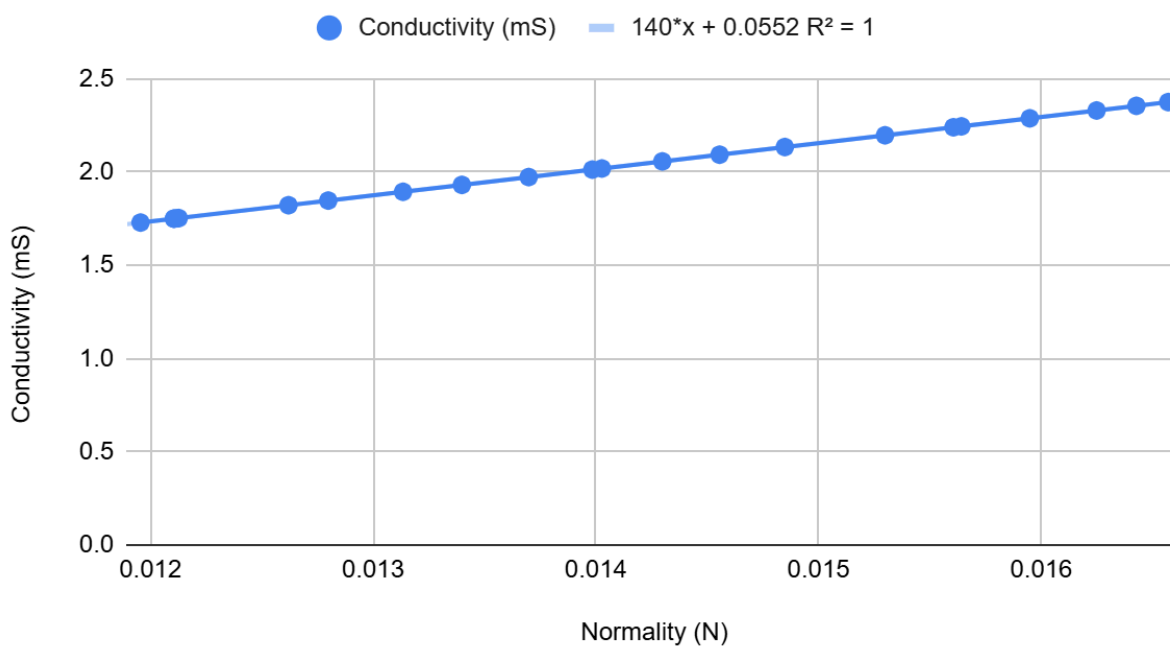


Figure 3: Figure showing the plot between Conductivity(mS) and Normality (N)

Concentration (M) of NaOH vs Time (s)

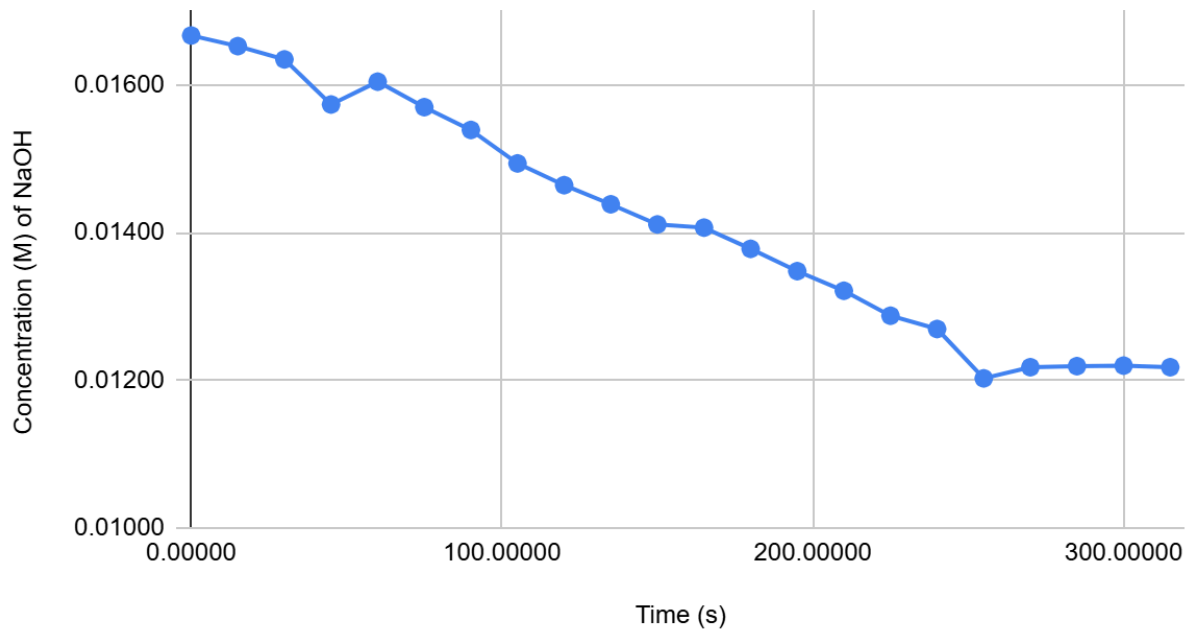


Figure 4: Figure showing the plot between Concentration (M) and Time (s)

Conversion (X_A) of NaOH vs. Time (s)

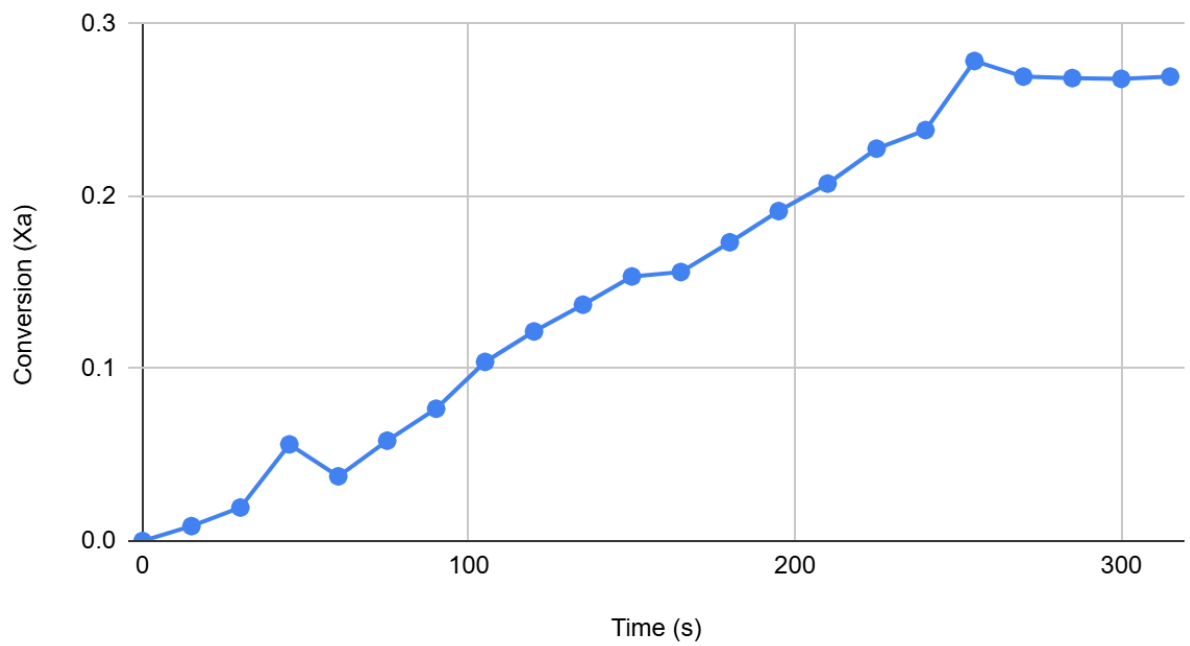


Figure 5: Figure showing the plot between Conversion (X_A) and Time (s)

From figure 1, we can see a curve having some equation. This curve is known as the calibration curve and the equation is known as the calibration equation. It is a curve between conductivity (mS) vs the Normality (M).

The calibration equation is : $y = 140 * x + 0.0552$

The Normality values for different conductivity values (taken in the interval of 15 s) from the reactor were calculated using the above obtained calibration equation.

The normality values obtained were then used to calculate the conversion. $X_A = 1 - \frac{N_{A0}}{N_A}$

From these values, we found that the highest conversion possible was 0.2694.

The conversion was then used to find the concentration, $C_A = C_{A0}(1 - X_A)$

Figure 2 shows the conductivity (mS) vs Time (s) curve. The conductivity is decreasing over time. Figure 3 shows the Conductivity (mS) vs Normality (N) plot, the conductivity is increasing with normality.

Explanation: In the saponification reaction, conductivity decreases over time as NaOH, a strong electrolyte, is consumed, reducing the number of free OH⁻ ions. The reaction products, sodium acetate and ethanol, contribute less to conductivity, further lowering it. Conversely, conductivity increases with normality because higher normality means a greater concentration of dissociated ions, enhancing charge mobility and electrical conduction.

Figure 4 shows the Concentration (M) vs Time (s) plot. The concentration of NaOH is decreasing. As the reaction proceeds, the NaOH concentration gradually declines, aligning with expected reaction kinetics. Initially, the decrease is rapid due to the high initial reactant concentration, driving a faster reaction rate. Over time, the rate of decline slows, indicating that the reaction is nearing equilibrium or steady-state conditions.

Figure 5 shows the Conversion vs Time (s) plot. The conversion of NaOH increases steadily over time, confirming its continuous consumption. Initially, the conversion rises rapidly before gradually stabilizing, indicating the reaction is approaching equilibrium. The theoretical conversion was 0.2564, while experimental values ranged from 0.23838 to 0.26942, showing good agreement with theoretical predictions.

The steady state in this experiment reached around 300 seconds as conductivity values became nearly constant.

Conclusion:

This experiment successfully analyzed the performance of a Continuous Stirred Tank Reactor (CSTR) for the second-order saponification reaction between sodium hydroxide (NaOH) and

ethyl acetate ($CH_3COOC_2H_5$) under steady-state conditions. Conductivity measurements were employed to monitor the reaction progress, providing a reliable calibration equation to determine NaOH concentration over time.

The results confirmed that NaOH concentration decreased steadily as the reaction proceeded, with conversion increasing until equilibrium was approached. The theoretical conversion (0.2564) was in good agreement with experimental values (0.23838 to 0.26942 when reaching steady state), validating the mass balance and kinetic models. The steady-state condition was achieved around 300 seconds, where conductivity values remained nearly constant.

Minor discrepancies in experimental and theoretical values were attributed to flow rate fluctuations, measurement uncertainties, and mixing inefficiencies. Despite these factors, the experiment effectively demonstrated the principles of CSTR operation, reinforcing its significance in industrial applications such as polymerization, wastewater treatment, and chemical synthesis, where precise reaction control is essential.

Sources of errors:

- There can be slight variations in the flow rate.
- There can be fluctuations in measuring conductivity.
- There can be human error while taking samples (we may not take samples every time at 15 seconds exactly).

References:

- [1]. O. Levenspiel, Chemical Reaction Engineering, 3rd ed. Wiley, 1999.
- [2]. J. Villiermaux, Chemical Engineering Approach to Reactor Design, 1st ed. Cambridge University Press, 1993.
- [3] 'Lab Manual' provided in the classroom.